

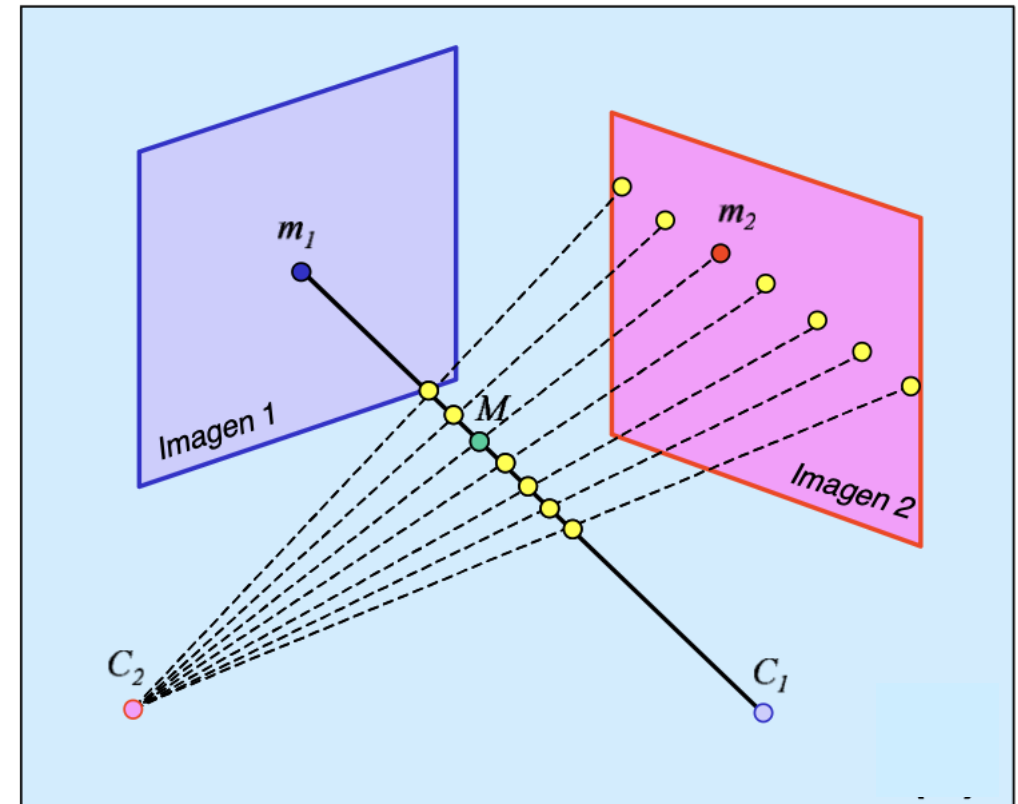


Geometría Epipolar

(CAPÍTULO 2 – CV02E)

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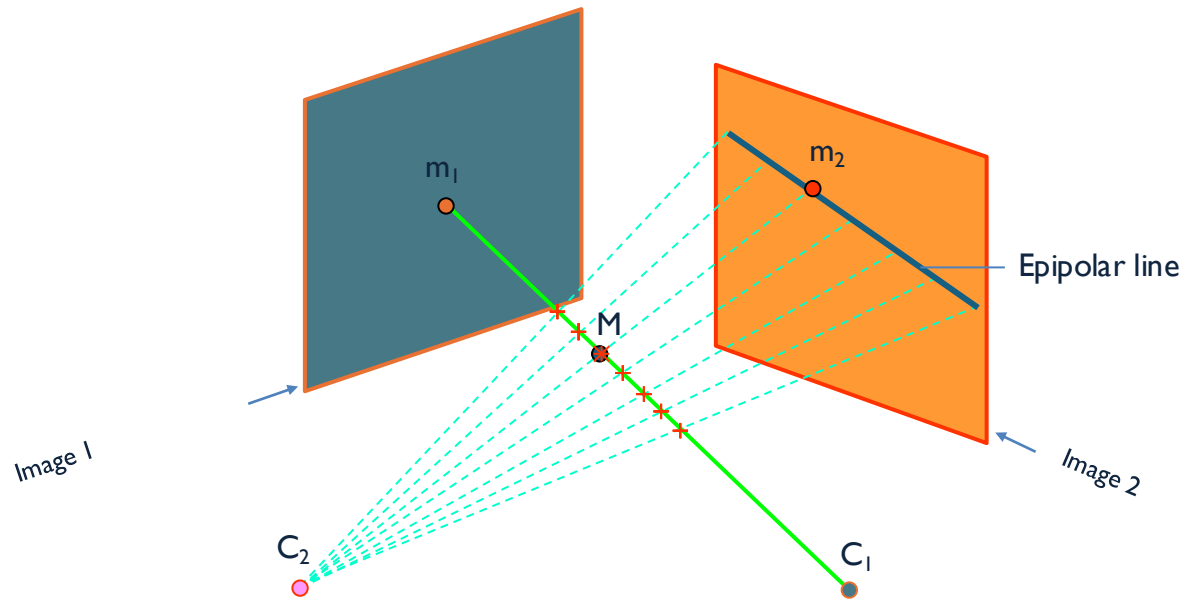


Correspondence in two images:

Two points are corresponding points if they are 2D projections of the same 3D point.



Epipolar Geometry:

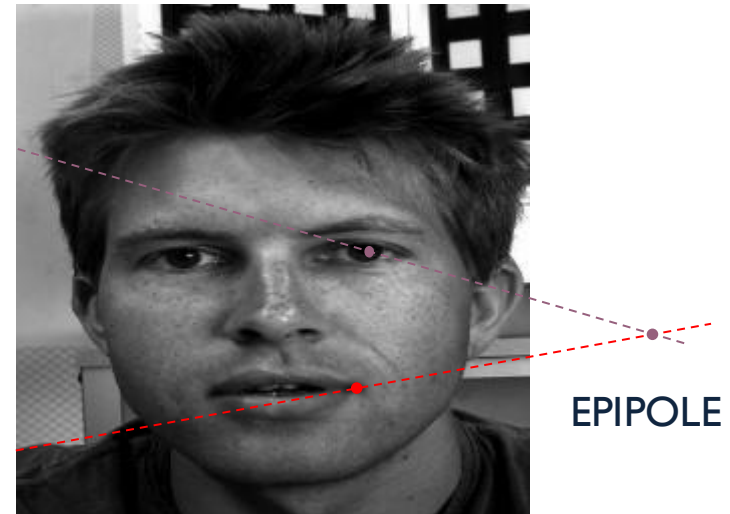


Epipolar Constraint:

If m_1 and m_2 are corresponding points
then m_2 lies on the epipolar line of m_1

Correspondence in two images:

Epipolar Constraint.



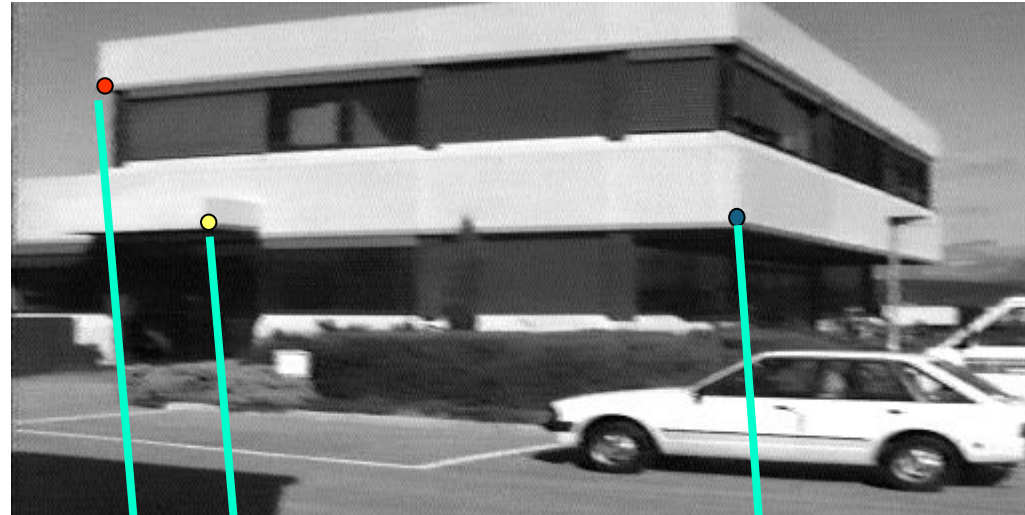


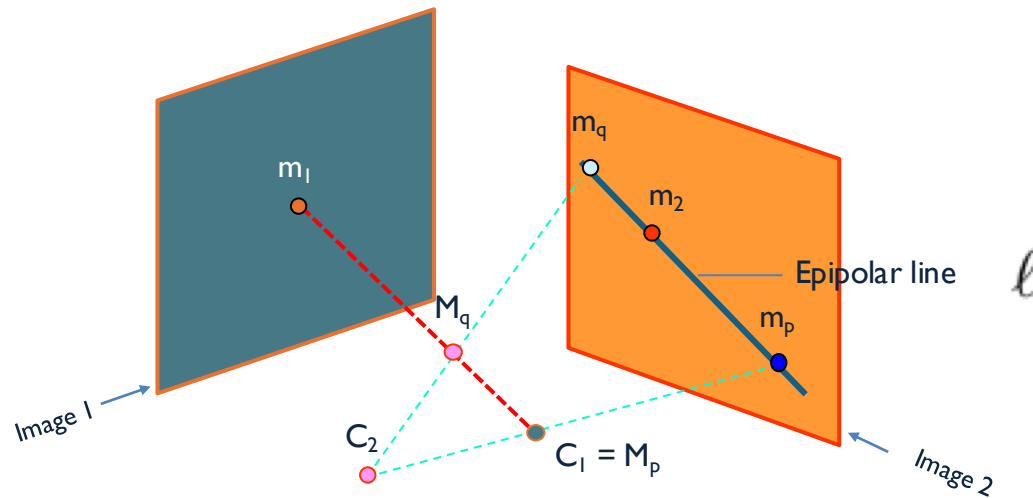
Image 1



Epipole

Image 2

Estimation of Fundamental Matrix:



[STEP 1]

2 points on $\langle C_1, m_1 \rangle$

$$1) M_p = C_1 = \text{null}(A)$$

$$2) M_q = A^T [A A^T]^{-1} m_1 = A^+ m_1$$

[STEP 2]

Projection onto image 2

$$\lambda_p m_p = B M_p, \quad \lambda_q m_q = B M_q$$

[STEP 3]

Epipolar Line

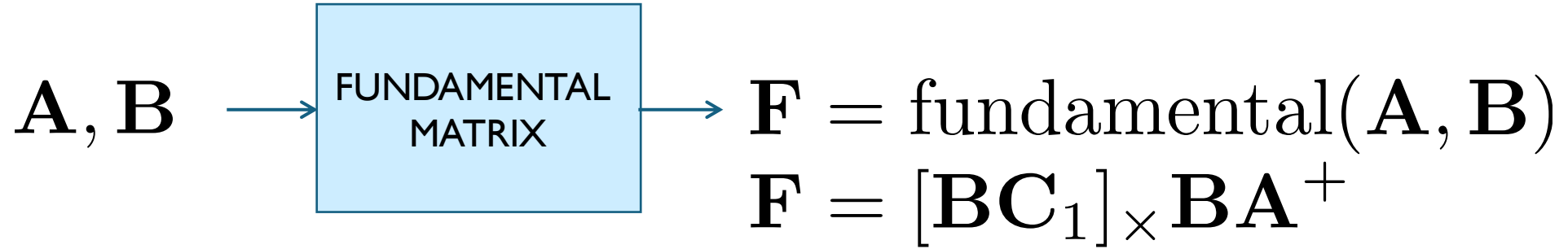
$$\ell = m_p \times m_q$$

$$= [B M_p] \times [B M_q]$$

$$= [B C_1]_x B A^+ m_1 = F m_1$$

Calibrated Views (A and B are known)

$$\begin{aligned}\lambda_1 \mathbf{m}_1 &= \mathbf{A} \mathbf{M} \\ \lambda_2 \mathbf{m}_2 &= \mathbf{B} \mathbf{M}\end{aligned}$$



where: $\mathbf{C}_1 = \text{null}(\mathbf{A})$

$$[\mathbf{u}]_{\times} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}$$

EPIPOLAR CONSTRAINT:

$$\mathbf{m}_2^T \mathbf{F} \mathbf{m}_1 = 0$$

$$\mathbf{u} \times \mathbf{v} = [\mathbf{u}]_{\times} \mathbf{v}$$

cross product

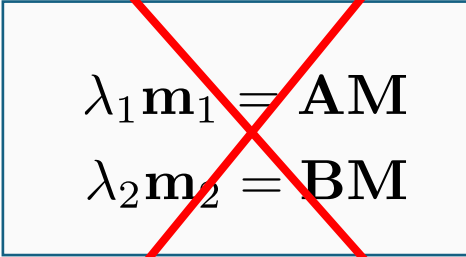
Practical Considerations

In practice, the projection points m_p and m_q can be corresponding points, if the perpendicular Euclidean distance from the epipolar line ℓ of the point m_p to the point m_q is smaller than a small number ε [12]:

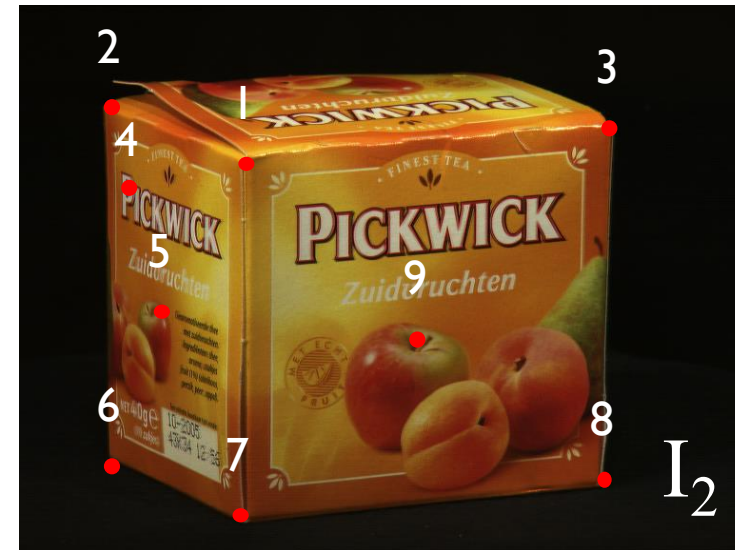
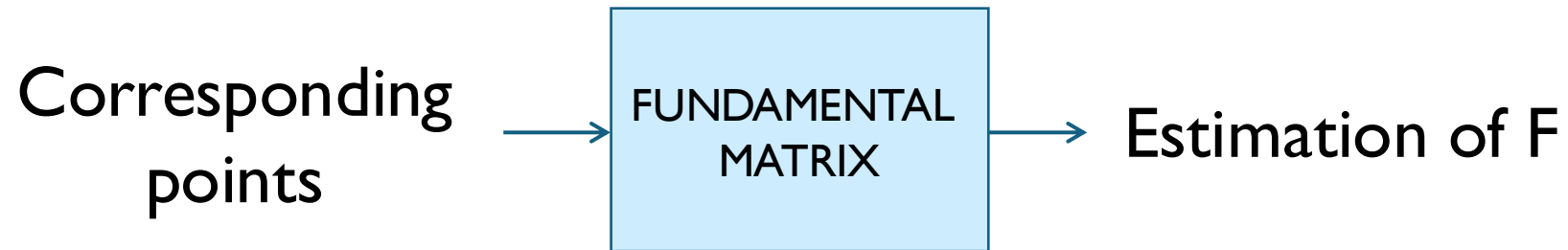
$$d_2 = \frac{|\mathbf{m}_q^T \mathbf{F}_{pq} \mathbf{m}_p|}{\sqrt{\ell_1^2 + \ell_2^2}} < \varepsilon, \quad (3.62)$$

where $\ell = \mathbf{F}_{pq} \mathbf{m}_p = [\ell_1 \ \ell_2 \ \ell_3]^T$.

Uncalibrated Views (A and B are unknown)


$$\lambda_1 \mathbf{m}_1 = \mathbf{A} \mathbf{M}$$
$$\lambda_2 \mathbf{m}_2 = \mathbf{B} \mathbf{M}$$

Uncalibrated Views (A and B are unknown)



For each pair of points $\mathbf{m}_1 = [x_1 \ y_1 \ 1]^\top$ and $\mathbf{m}_2 = [x_2 \ y_2 \ 1]^\top$
 we have the epipolar constraint: $\mathbf{m}_2^\top \mathbf{F} \mathbf{m}_1 = 0$

$$\begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix} \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix} \begin{bmatrix} F_{11}x_1 + F_{12}y_1 + F_{13} \\ F_{21}x_1 + F_{22}y_1 + F_{23} \\ F_{31}x_1 + F_{32}y_1 + F_{33} \end{bmatrix} = 0$$

$$(F_{11}x_1 + F_{12}y_1 + F_{13})x_2 + (F_{21}x_1 + F_{22}y_1 + F_{23})y_2 + (F_{31}x_1 + F_{32}y_1 + F_{33}) = 0$$

$$F_{11}x_1x_2 + F_{12}y_1x_2 + F_{13}x_2 + F_{21}x_1y_2 + F_{22}y_1y_2 + F_{23}y_2 + F_{31}x_1 + F_{32}y_1 + F_{33} = 0$$

For each pair of points $\mathbf{m}_1 = [x_1 \ y_1 \ 1]^\top$ and $\mathbf{m}_2 = [x_2 \ y_2 \ 1]^\top$
 we have the epipolar constraint: $\mathbf{m}_2^\top \mathbf{F} \mathbf{m}_1 = 0$

$$\begin{bmatrix} x_1 x_2 & y_1 x_2 & x_2 & x_1 y_2 & y_1 y_2 & y_2 & x_1 & y_1 & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

$$F_{11}x_1x_2 + F_{12}y_1x_2 + F_{13}x_2 + F_{21}x_1y_2 + F_{22}y_1y_2 + F_{23}y_2 + F_{31}x_1 + F_{32}y_1 + F_{33} = 0$$

For each pair of points $\mathbf{m}_1 = [x_1 \ y_1 \ 1]^\top$ and $\mathbf{m}_2 = [x_2 \ y_2 \ 1]^\top$ ($i = 1 \dots n$)
 we have the epipolar constraint: $\mathbf{m}_2^\top \mathbf{F} \mathbf{m}_1 = 0$

$$\begin{bmatrix} x_1^1 x_2^1 & y_1^1 x_2^1 & x_2^1 & x_1^1 y_2^1 & y_1^1 y_2^1 & y_2^1 & x_1^1 & y_1^1 & 1 \\ x_1^2 x_2^2 & y_1^2 x_2^2 & x_2^2 & x_1^2 y_2^2 & y_1^2 y_2^2 & y_2^2 & x_1^2 & y_1^2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1^n x_2^n & y_1^n x_2^n & x_2^n & x_1^n y_2^n & y_1^n y_2^n & y_2^n & x_1^n & y_1^n & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

We can set $F_{33} = 1$

(because $\mathbf{F}$ is homogeneous)

For each pair of points $\mathbf{m}_1 = [x_1 \ y_1 \ 1]^\top$ and $\mathbf{m}_2 = [x_2 \ y_2 \ 1]^\top$ ($i = 1 \dots n$)
 we have the epipolar constraint: $\mathbf{m}_2^\top \mathbf{F} \mathbf{m}_1 = 0$

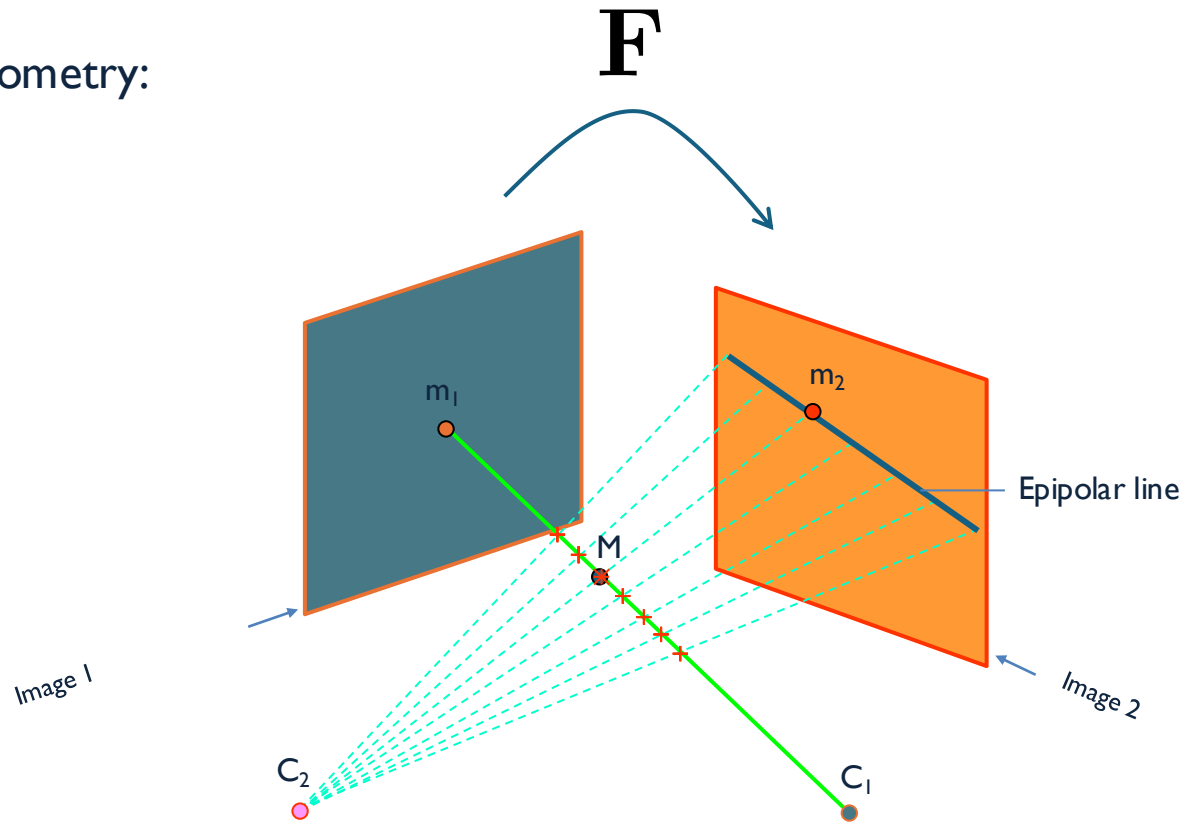
with $F_{33} = 1$

$$\begin{bmatrix} x_1^1 x_2^1 & y_1^1 x_2^1 & x_2^1 & x_1^1 y_2^1 & y_1^1 y_2^1 & y_2^1 & x_1^1 & y_1^1 \\ x_1^2 x_2^2 & y_1^2 x_2^2 & x_2^2 & x_1^2 y_2^2 & y_1^2 y_2^2 & y_2^2 & x_1^2 & y_1^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1^n x_2^n & y_1^n x_2^n & x_2^n & x_1^n y_2^n & y_1^n y_2^n & y_2^n & x_1^n & y_1^n \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}$$

$$\mathbf{QF}' = \mathbf{r} \quad \mathbf{F}' = \mathbf{Q}^{-1} \mathbf{r} \quad \text{for } n=8$$

$$\mathbf{F}' = [\mathbf{Q}^\top \mathbf{Q}]^{-1} \mathbf{Q}^\top \mathbf{r} \quad \text{for } n>8$$

Epipolar Geometry:



$$\ell = \mathbf{F}\mathbf{m}_1$$

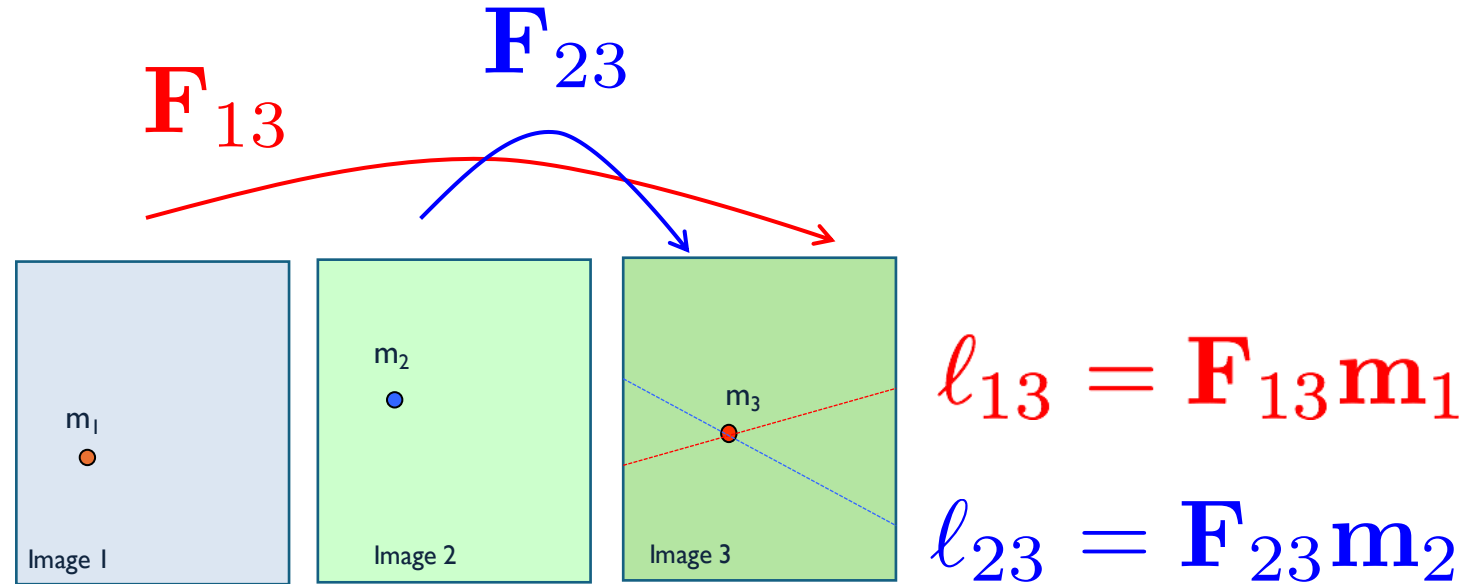
$$\mathbf{m}_2 \cdot \ell = 0$$

Epipolar Constraint:

If $\mathbf{m}_1$ y $\mathbf{m}_2$ are corresponding points
then $\mathbf{m}_2$ lies on the epipolar line of $\mathbf{m}_1$

$$\mathbf{m}_2 \cdot \ell = \mathbf{m}_2^T \ell = \mathbf{m}_2^T \mathbf{F} \mathbf{m}_1 = 0$$

Epipolar Geometry for 3 views:



$$\mathbf{F}_{13} = \text{fundamental}(\mathbf{A}, \mathbf{C})$$

$$\mathbf{F}_{23} = \text{fundamental}(\mathbf{B}, \mathbf{C})$$

Problem:

Given $\mathbf{m}_1, \mathbf{m}_2$ (and projection matrices $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$)
find $\mathbf{m}_3$

$$\mathbf{m}_3 = \ell_{13} \times \ell_{23}$$

$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_3 \mathbf{m}_3 = \mathbf{C}\mathbf{M}$$